

Debrup Das

📍 Amherst, MA ✉️ debrupdas@umass.edu in [DebrupDas](#)

Education

- Ph.D.** *Computer Science, University of Massachusetts, Amherst, MA, USA* Sept 2024 – Ongoing
- **Advisor:** [Prof. Negin Rahimi](#)
 - **Member of CIIR Lab** ([Center for Intelligent Information Retrieval](#)), CIIR Director: [Prof. James Allan](#)
 - **Anticipated Graduation Date:** September 2029
- M.Sc.** *Mathematics and Computing, Indian Institute of Technology Kharagpur, India* Aug 2019 - Apr 2024
[Integrated Master of Science 5 yr]
- **Thesis Advisor:** [Prof. Somak Aditya](#)
 - **CGPA:** 9.00/10

Research Interests

My research focuses on building retrievers that reason about relevance rather than matching surface similarity. I am currently working on: (1) *Scaling test-time thinking for reasoning retrieval*, (2) *Reinforcement learning approaches to train retrievers optimized under embedding objectives and retrieval rewards*, (3) *Memory-augmented systems that retrieve and continually learn from past experiences*.

Current Research

- Thinking Reasoning Retrievers, UMass Amherst** Research Assistant
May 2026 – Current
- Research on developing strong retrievers capable of thinking, through new RL training approaches, guided by rewards specifically designed for retrieval.
 - Mentored by [Prof Negin Rahimi](#).
- Training Dense Retriever Models for Reasoning Retrieval, UMass Amherst** Research Assistant
Sep 2024 – Apr 2025
- Research on developing reasoning-intensive retrievers from raw mathematical data (without any notion of document relevance), which shows strong generalizability to diverse domains.
 - Mentored by [Prof Negin Rahimi](#).
 - Accepted as Main Conference Paper at **EMNLP 2025**, Suzhou, China.

Industry Research Internships

- Samsung Research America (SRA), AI Center – Mountain View** Research Intern
Feb 2026 – May 2026
- Research on building **latent-memory-augmented systems** for continual learning of long-context personalized user memory.

- Encoded user history as latent memory slots projected into **dynamic soft prompts** at inference, beating LoRA by 8.8% on PersonaMem v1 at **64× lower KV-cache cost**.
- Accepted at **EMNLP 2026**, Budapest. Mentored by **Srinivas Chappidi** [🔗](#).

Rakuten, Language and Speech Processing Team

Research Intern
May 2023 – Dec 2023

- Research on *tool-augmented LLMs* for complex math reasoning.
- Integrating LLMs with symbolic tools such as *Python*, *Wolfram Alpha* and web search with *Bing API*.
- Accepted as a Main Conference Paper at **NAACL 2024**, Mexico City.

Academic Research Internships

Indian Institute of Technology, Kharagpur (in collaboration with **MBZUAI**)

Research Intern
Dec 2023 – May 2024

- Research on *sensitivity* of LLMs to different segments of the input, in classification tasks such as hate speech, sentiment analysis, etc.
- *Adversarial example generation* and jailbreak attacks using the notion of sensitivity. Mentored by **Prof Monojit Choudhury** [🔗](#) and **Prof Somak Aditya** [🔗](#), funded by **Microsoft AFMR** [🔗](#).
- Accepted as a Main Conference Paper at **NAACL 2025**, Albuquerque, New Mexico.

McGill University, Montreal, Genome Quebec Innovation Center

Research Intern
May 2022 – Dec 2022

- Genome-wide association studies (GWAS) to detect rare disease-causing variants, mentored by **Prof Simon Gravel** [🔗](#).
- Funded by the *MITACS Globalink Research Internship Program*.

Indian Institute of Technology, Kharagpur

Research Intern
May 2021 – Dec 2021

- Training tree-based deep learning models (*Genetic programming*) with nature-inspired optimization methods such as evolutionary algorithms. Mentored by **Prof Nirupam Chakraborti**, **Czech Technical University** [🔗](#).

Publications

Latent Personal Memory: Represent Personal Memory as Dynamic Soft Prompts

Nov 2026

Debrup Das, Avinash Amballa, Yashas Malur Saidutta, Vijay Srinivasan, Vivek Kulkarni, Srinivas Chappidi

EMNLP 2026 Findings [🔗](#)

RaDeR: Reasoning-aware Dense Retrieval Models

Nov 2025

Debrup Das, Sam O’Nuallain, Negin Rahimi

EMNLP 2025 Main Conference [Website](#)

MathSensei: A Tool-Augmented Large Language Model for Mathematical Reasoning

Jun 2024

Debrup Das, Debopriyo Banerjee, Somak Aditya, Ashish Kulkarni

NAACL 2024 Main Conference [Featured on Rakuten News](#)

SMAB: MAB based Word Sensitivity Estimation Framework and its Applications in Adversarial Text Generation

Apr 2025

Saurabh Kumar Pandey, Sachin Vashistha, **Debrup Das**, Somak Aditya, Monojit Choudhury

NAACL 2025 Main Conference [Website](#)

Languages and Technologies

Languages: Python, Bash, R, SQL, C, C++

Libraries: PyTorch, Transformers, FSDP, DeepSpeed, VeRL, NumPy, Scikit-Learn, SciPy, SymPy

Tools: Git, $\text{\LaTeX}$, HuggingFace, Wolfram Alpha

Teaching

- Teaching Assistant, UMass Amherst, Fall 2026, **Information Retrieval** (CS 646) by [Prof Negin Rahimi](#)
- Teaching Assistant, UMass Amherst, Summer 2025 (Online), **Practice and Applications of Data Management** (CS 345) by [Prof Gordon Anderson](#)
- Teaching Assistant, UMass Amherst, Spring 2025, **Search Engines** (CS 446) by [Prof Negin Rahimi](#)
- Teaching Assistant, UMass Amherst, Fall 2024, **Search Engines** (CS 446) by [Prof James Allan](#)

Service

- Reviewer, **ACM Transactions on Information Systems (TOIS)**.
- Reviewer, **North American Chapter of Association for Computational Linguistics NAACL 2025**

English Proficiency

TOEFL: TOTAL: 112/120, SPEAKING: 30/30, READING: 29/30, LISTENING: 27/30, WRITING: 26/30